

Validation of Medical Imaging AI

Practical Workbook and Solutions

Part I: Classification Models

Practical 1.1: Accuracy versus Balanced Accuracy

Accuracy = 97%, Sensitivity = 0, Specificity = 1, Balanced Accuracy = 0.5

How would you judge the model based on Accuracy alone?

Based solely on Accuracy, the model appears excellent because 97% is a high value. However, this conclusion is misleading because the dataset is highly imbalanced.

Would this model be acceptable for a screening task? Why or why not?

No. A screening model should detect as many diseased patients as possible. This model misses every diseased patient.

In medical AI, which type of error may be more dangerous in a screening context?

Depends. However, in many screening settings false negatives are considered more dangerous because diseased patients may remain undetected and untreated.

Why does Accuracy look high while Balanced Accuracy is much lower?

Accuracy is dominated by the large number of healthy patients. Balanced Accuracy evaluates performance separately for healthy and diseased patients and therefore reveals the complete failure on the diseased class.

Practical 1.2: Accuracy, PPV and Disease Prevalence

Prevalence = 0.02%, PPV = 0.30%, Accuracy = 95.0%, Sensitivity = 75.0%, Specificity = 95.0%, Balanced Accuracy = 85.0%

Which metric is most affected by the large number of false positives?

Metric	Effect of increasing FP
Accuracy	Moderate decrease
Balanced Accuracy	Small decrease
Sensitivity	No effect
Specificity	Moderate decrease
PPV	Very large decrease

Which metric is most affected by the rarity of positive cases?

Metric	Dependence on prevalence
Accuracy	Moderate
Balanced Accuracy	Low
Sensitivity	Minimal
Specificity	Minimal
PPV	Very high

Part II: Segmentation Models

Practical 2: Dice Similarity Coefficient (DSC)

Case	Prediction	DSC
Case 1	Prediction 1	0.98
Case 1	Prediction 2	0.95
Case 2	Prediction 1	0.86
Case 2	Prediction 2	0.40

How would you compare the two predictions across cases?

The two cases differ mainly in the size of the target structure. In both cases, Prediction 1 misses one foreground pixel compared to the reference, while Prediction 2 misses two foreground pixels. The absolute segmentation errors are therefore similar across cases.

How does the DSC change between the different predictions?

For both cases, DSC decreases from Prediction 1 to Prediction 2 because Prediction 2 contains a larger segmentation error. However, the effect is much stronger for Case 2. While the DSC changes only slightly for Case 1, the same increase in error leads to a substantial decrease in DSC for Case 2.

Why do you think the DSC behaves differently across the two cases?

The DSC measures overlap between reference and prediction relative to the size of the segmented structure. Although the absolute error is similar in both cases, the missing pixels represent a much larger proportion of the structure in Case 2 than in Case 1. Consequently, the overlap decreases more strongly relative to the structure size, resulting in a larger reduction in DSC. This example illustrates that overlap-based metrics such as DSC (and IoU) can be particularly sensitive to errors in small structures.